

Blockchain-based Digital Identity for Secure Authentication of IoT Devices In 5G Networks

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Abstract -The growing use of IoT in 5G networks has led to new security concerns, particularly related to the lack of authentication mechanisms. This paper proposes a new way to solve this problem by using blockchain-based digital tokens to identify IoT devices and enable secure communication over 5G networks, etc. Our Digital Identity Authentication using Blockchain (DIA-BC) approach proposed in this paper is to create a unique blockchain- based digital identity for each device. The created digital identities are verified using cryptographic protocols and stored in a tamper- proof decentralized ledger. These digital identities can be used to authenticate devices as they connect to a network, preventing unauthorized access, and allowing secure communication channels to be established. We discuss the benefits of this approach, including improved security, reliability, and deployment, and highlight some of the challenges that must be addressed when implementing this approach. Our plan shows our commitment to improving the security of IoT devices in 5G networks, with the ability to create safe and secure IoT devices that can share communications with each other without a network.

Keywords: 5G networks, Blockchain technology, Digital identities, IoT, Authentication, Security, Confidentiality.

I. INTRODUCTION

The surge in IoT devices demands robust security measures, amplified by 5G networks' rapid integration. Blockchain technology emerges as a solution, crafting decentralized tokens for IoT devices. These tokens establish transparent and tamper-proof identity management, restricting network access to authorized devices. Blockchain authentication enhances data trustworthiness, ensuring traceability to the original IoT device. The distributed nature of IoT with blockchain fortifies security, resists interference and fraud, offering cost-effective trust and increased freedom [1][2]. This article explores blockchain's potential in securing IoT within 5G networks, addressing challenges and emphasizing digital identity management benefits. Practical implementation includes smart contract creation and seamless integration with 5G networks.

II. RELATED WORK

Raseena et al. [3] propose blockchain integration into 5G IoT, addressing and providing solutions for security concerns, especially in heterogeneity, latency, scalability, and overall security. M.F.K. Sial [4] highlights key security issues in 5G IoT, suggesting that blockchain can effectively address challenges such as heterogeneity, latency, scalability, and privacy. The evolution of 5G networks is poised to significantly enhance current IoT capabilities, revolutionizing cellular operations, addressing IoT security concerns, and tackling network challenges. This advancement is propelling the future of the Internet towards an edge-centric paradigm [5]. Rodriguez B et al. [6] introduce innovative models using smart contracts and blockchain for adaptable DDoS mitigation globally. Alexopoulos et al. [7] emphasize the benefits of open-source documents (ODL) and blockchain in trust management, surpassing traditional approaches. Dongxing Li et al. [8] address criticisms of traditional identity identification in IoT, presenting a new blockchain-based IoT authentication paradigm using Hyperledger Fabric. Munkenyi Mukhadi et al. [9] present a consensus-based blockchain for IoT devices, relying on identity-authenticated light sources and a secure, unchangeable ledger. Chae, CJ., and Cho, HJ [10] propose an authentication system combining session and public keys, with certificate sharing managed by a central authentication server. The contrasting findings of [11] and [12] involve IoT device detection using fingerprinting techniques, proven ineffective for unidentified equipment [13]. The proposed hypothesis in this literature offers a comprehensive approach, providing more work and potential solutions in contrast to existing strategies. Overall, these studies collectively showcase the potential of blockchain in addressing diverse security challenges within the evolving landscape of 5G IoT. In [14], identity-based P2P authentication with blockchain is proposed to address the single point of

failure issue associated with Certificate Authority (CA) servers in IoT devices. The authentication relies on public and private keys, utilizing the blockchain validator node sub chain for verification. However, the method's dependence on a reliable authenticator raises concerns, as the authenticator node intervenes only if the sender is rejected, potentially exposing vulnerabilities to malicious nodes. Contrastingly, the Digital Identity Authentication using Blockchain (DIA-BC) method offers secure communication and IoT device authentication across 5G networks. This approach enhances security through cryptographic protocols and blockchain, ensuring impenetrable and safe authentication for sensitive data transfer, preventing unauthorized access, and maintaining data privacy. The distributed ledger eliminates the need for centralized network management, contributing to resilience and the development of a trustless environment. Additionally, the DIA-BC method prioritizes reliability by reducing errors and system breakdowns through digital identity authentication before network access, crucial for continuous uptime in IoT applications. Its scalability accommodates numerous devices and transactions in large-scale IoT networks, enabled by decentralized ledgers and blockchain technology. Furthermore, auditability, transparency, and immutability are ensured, guaranteeing document integrity and reducing the likelihood of fraud. Users benefit from a decentralized ledger, allowing them to track transactions and verify information, reinforcing transparency and auditability.

III. PROPOSED APPROACH

Developing self-defence mechanisms is crucial for security, given that attackers may gain control and manipulate authenticated devices even after the initial authentication [9]. Combining Public

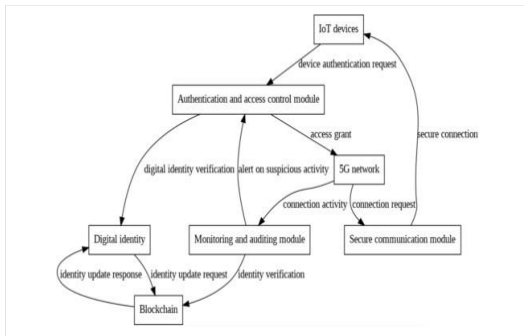


Fig 1: Block diagram of Digital Identity Authentication using Blockchain (DIA-BC).

Key Infrastructure (PKI) with blockchain technology provides a viable solution to security and privacy concerns in the expanding realm of internet-connected devices, especially in 5G networks.

Fig 1 illustrates a digital identity system creating unique IDs for each registered IoT device, storing them securely on the blockchain. The developed Protocol, a fusion of PKI and blockchain, establishes secure communication protocols

leveraging IoT device digital identities for authenticity verification and ensuring tamper-proof data transfers. Blockchain technology stores and validates digital signatures for each device, while the PKI protocol encrypts and decodes data transmissions, ensuring a robust security framework. The implemented secure communication protocols and digital identification systems in 5G networks prioritize scalability, dependability, and security. The integration of blockchain networks with 5G infrastructure enhances the overall security of software applications facilitating interaction with blockchains for authenticating and validating IoT devices. This comprehensive approach addresses the evolving challenges in securing IoT devices within the dynamic landscape of 5G networks. Test systems validate the reliability and effectiveness of the developed secure communication protocols, ensuring the protection of IoT devices from potential threats. The collaborative use of PKI and blockchain technology establishes a resilient security foundation, crucial for fostering trust and safeguarding data integrity in the interconnected ecosystem of IoT devices.

IV. ARCHITECTURE

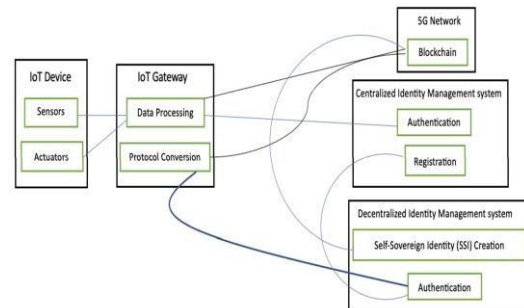


Fig 2: Architecture of DIA-BC.

IoT Devices authenticate through the Digital Identity Authentication using Blockchain (DIA-BC) system, depicted in Figure 2. The Identity Management System (IMS) is responsible for creating and managing digital identities for IoT devices, storing unique identifiers, authentication keys, and access control policies. IMS interacts with the blockchain network, facilitating the storage and retrieval of device identity information securely. The Blockchain network serves as the core component of DIA-BC, providing a decentralized platform for managing and securing device identities. Smart Contracts within DIA-BC act as stand-alone agreements, executing IoT device authentication, activating access control policies, and validating personal data on devices. Leveraging the capabilities of 5G networks, DIA-BC enables rapid and reliable communication among IoT devices and physical objects. 5G

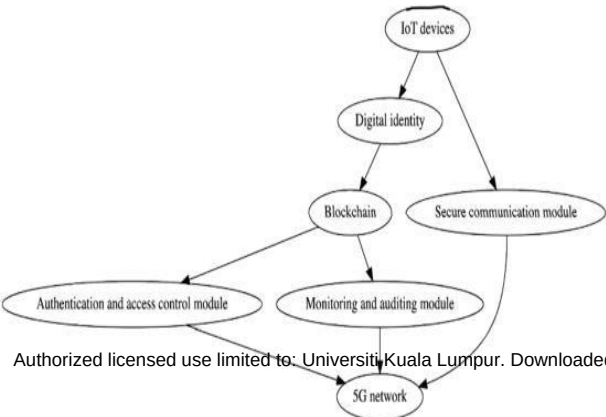
networks are particularly suitable for IoT applications, offering high-bandwidth connections with low latency. Security Management oversees DIA-BC's security, implementing safeguards against illegal data access and modification, including data integrity checks and access encryption. In 5G networks, the design ensures a secure platform for detecting IoT devices, limiting network access to authorized devices. DIA-BC ensures decentralized and transparent asset self-management through blockchain networks and smart contracts, reinforcing security and efficiency in IoT environments.

V. DESIGN

Architectural design is crucial, defining components like blockchain networks, digital identification systems, smart contracts, and monitoring modules based on requirements analysis. Software development follows, involving programming smart contracts, creating digital identity systems, and developing monitoring and auditing modules. Hardware implementation is essential, requiring the setup of blockchain networks, deployment of smart contracts, and the establishment of communication between IoT devices and gateways. Extensive testing and debugging ensure the system functions as intended, incorporating monitoring, review standards, fact-checking, and access control protocols. Deployment for production use follows successful testing, emphasizing ongoing maintenance for system security and updates. The system architecture is developed based on requirements analysis, outlining key components like blockchain networks, digital identification systems, smart contracts, and monitoring modules. Software development involves programming smart contracts, creating digital identity systems, and developing monitoring and auditing modules. Hardware implementation requires setting up blockchain networks, deploying smart contracts, and establishing communication between IoT devices and gateways.

VI. ALGORITHM

The Practical Byzantine Fault Tolerance



(PBFT) algorithm was designed to reduce latency and enhance throughput in consensus mechanisms [15]. Specifically tailored for permissioned blockchain networks with limited nodes requiring network identification and authentication, PBFT is implemented. PBFT operates on a system of rounds, wherein each round involves a sequence of steps executed by participating nodes. This algorithm aims to ensure consensus and fault tolerance in permissioned blockchain networks, optimizing their efficiency and reliability.

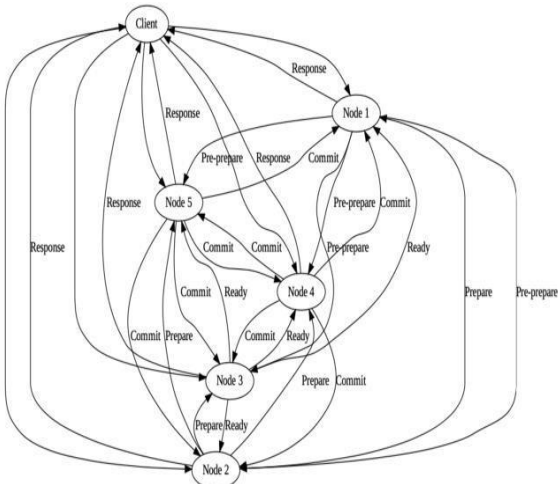


Fig 3: Workflow model of PBFT algorithm.

The client initiates a request in the first stage, disseminating it to all nodes on the network (Figure 3). In the second phase, each node verifies the request and signals readiness by sending a ready message to all other nodes. Every node, in the third step, receives ready messages from all others and responds with its own ready message, indicating preparedness for the next phase. The fourth step involves nodes receiving prepare messages, sending commit messages, and signaling readiness for the subsequent step. In the final step, each node executes the request upon receiving commit messages from all others, returning the response to the client. If more than two-thirds of nodes send commit messages, the request is considered committed.

VII. WORKING METHODOLOGY

Our DIA-BC relies on Public Key Infrastructure (PKI) for secure digital certificate issuance and management, utilizing the Hyperledger Fabric framework. The consensus mechanism combines Proof of Work (PoW) and Practical

Byzantine Fault Tolerance (PBFT) to enhance throughput and reduce latency.

Fig 4: Flow model of the DIA-BC.

The Secure Hash Algorithm (SHA-256) ensures blockchain data integrity, while the Elliptic Curve Digital Signature Algorithm (ECDSA) generates and validates digital signatures. In device registration, unique digital identities are created on the blockchain, ensuring tamper-proof, cryptographically secure records, including the device's public key. Authentication involves devices using private keys to sign digital messages, with the network verifying the signature against the public key on the blockchain. Access control policies are then applied based on the device's digital identity, ensuring appropriate resource and service access. Secure communication is established using encryption protocols like SSL/TLS, protecting data exchange. Continuous monitoring and auditing detect security threats, such as unusual access attempts or excessive data transfers, ensuring a robust security framework. This recommended approach, with blockchain-based digital identities, enhances the security and reliability of IoT devices on 5G networks, safeguarding authentication and communication through encryption and decentralized blockchain structures.

VIII. IMPLEMENTATION

This approach involves creating unique, tamper-proof blockchain-based digital identities for each IoT device on the 5G network, verified through cryptographic protocols. During authentication, devices use private keys to sign messages, validated by the public key stored on the blockchain. Once authenticated, devices access the network securely, preventing unauthorized connections. In low-power systems, integrating Public Key Infrastructure (PKI) involves modifications like smaller key sizes and ECC emphasis for efficiency. The use of blockchain-based digital tokens enhances security on 5G networks by preventing forgery, creating a decentralized, unified IoT system without the need for a central controller.

IX. COMPARATIVE STUDY

Our work addresses vital security concerns for IoT devices and 5G networks by offering tamper-proof personal data and decentralized solutions. This approach enhances overall security, eliminating the risk of a single point of failure and increasing system resilience. The solutions guarantee secure data transmission, preserving data integrity, and provide effective access control management for authorized users on IoT devices and 5G networks. The proposed Decentralized Identity and Access Management with

Blockchain (DIA-BC) mechanism introduces a decentralized, blockchain-based approach to identity management, ensuring a high level of security through robust encryption and blockchain technology. Its distributed nature allows for high scalability, and the use of lightweight consensus algorithms and smart contracts ensures efficient performance. This solution comes at a moderate cost, balancing security with infrastructure and development expenses. In comparison, OAuth 2.0 and OpenID Connect prioritize simplicity and wide adoption with centralized token-based authentication, compromising security to some extent. SAML and XACML excel in security with encryption and digital signatures but incur higher costs due to their complexity and infrastructure requirements. LDAP provides a moderate level of security with highly scalable directory-based architecture and efficient performance, all at a moderate cost. RADIUS emphasizes simplicity and low cost but sacrifices some security. Kerberos and Certificates offer high security with centralized elements, impacting scalability and incurring higher costs due to complexity and infrastructure requirements [16]–[24]. Traditional methods evaluated in the study include password authentication, biometric authentication, smart card authentication, and public key infrastructure (PKI). The proposed blockchain-based method is superior to existing methods in terms of security, scalability, and cost. However, the performance is slightly lower, but the difference is small. The promise of blockchain-based digital identities for safe authentication and IoT device access control in 5G networks is demonstrated by this study. This further demonstrates the requirement for scalable security methods and procedures to handle the expansion of linked devices in Internet of Things networks.

X. SIMULATION SETUP

The simulation of a 5G network, IoT devices, blockchain chain, network layer, and events was conducted using Google Colab and SimPy, a powerful discrete-event simulation library for complex systems in Python. SimPy facilitated modeling system behavior, including IoT devices, 5G networks, blockchain-based digital identity, networking layer, and event interactions. Key parameters for IoT devices, 5G networks, and blockchain were defined, and Pyblockchain, Socket, and SimPy packages were imported. The simulation considered factors such as

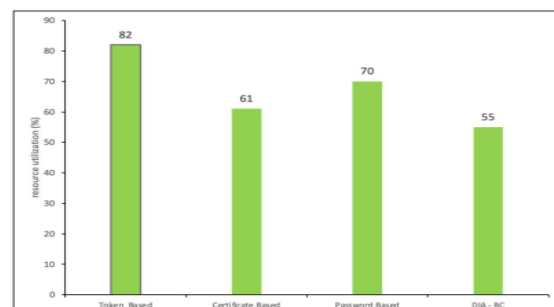


Fig 6: Resource utilization in terms of memory and CPU

the number of devices, traffic type, latency, bandwidth, and network consensus for blockchain. The study explored network latency, bandwidth, protocol, nodes, and SimPy events for predicting IoT device and vehicle management arrival behavior. The simulation characteristics, such as IoT device number and network connection model, were specified. The network protocol for IoT device communication and blockchain-based identity verification was implemented using the sockets library.

XI. CONTRIBUTION

DIA-BC employs blockchain for secure IoT device management in 5G networks, ensuring authentic and immutable personal data. Leveraging cryptographic blockchain tech, it enhances authentication efficiency and data authorization security. The system ensures swift transaction processing through consensus algorithms, reducing latency and boosting throughput. DIA-BC's scalability allows seamless addition of nodes and

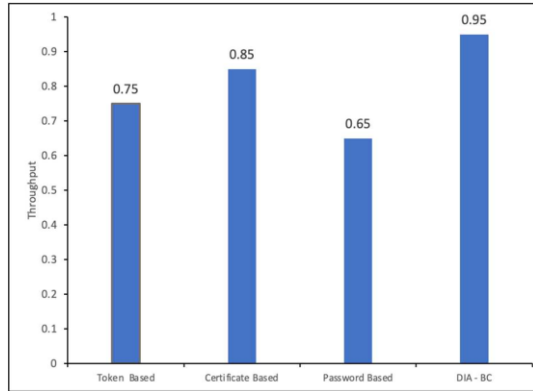


Fig 5: Throughput analysis.

devices without compromising security or performance.

In Fig 5, the bar chart illustrates throughput scores, with our proposed authentication method achieving the highest score of 0.95. DIA-BC outperforms other methods by streamlining authentication steps and optimizing resource utilization, leading to enhanced efficiency.

In Fig 6, CPU and memory utilization are measured metrics, depicted by bars for each method. The bar height signifies resource utilization levels, while the X-axis indicates the authentication methods, and the Y-axis represents utilization as a percentage.

The proposed work demonstrates that blockchain-based digital identity systems enhance secure authentication for IoT devices in 5G networks, surpassing other methods in performance and scalability. DIA-BC not only ensures privacy but also empowers users with control over their personal data, contributing significantly to the advancement of secure authentication in the 5G ecosystem.

XII. CONCLUSION

This article proposes a blockchain-based solution for securing and managing IoT devices in 5G networks, providing a distributed and secure digital identity system. The system outperforms various existing authentication methods in terms of security, scalability, and performance. Simulation results confirm its feasibility and efficiency. The study suggests avenues for future exploration, including the integration of blockchain and artificial intelligence for enhanced security. Drawing from a literature review, the article offers valuable insights for academics, developers, and legislators seeking to improve IoT device security in 5G networks, ultimately enhancing network security and access control capabilities.

XIII. FUTURE WORK

To advance blockchain-based digital identity systems for IoT devices in 5G networks, future research should explore new blockchain technologies, enhance PKI systems, and devise methods for securing data transfers. Integration of artificial intelligence can enhance security by detecting anomalous device behavior and customizing access control policies. Utilizing advanced blockchain technologies like sharding and sidechains can improve scalability and transaction times, accommodating more IoT devices. Further, adopting decentralized identity solutions like Decentralized Identifiers (DID) offers a flexible and trust-enhancing approach, allowing users to control their information.

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